

UJIAN TENGAH SEMESTER

Pemrograman Backend Lanjut Praktikum

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Link repo GitHub: <https://github.com/finarch1ve/api-students/tree/main/UTS>

Deskripsi

SIKAD Mini API adalah RESTful API back end layanan akademik sederhana yang mengelola data mahasiswa, mata kuliah, dan Kartu Rencana Studi (KRS) untuk dua peran, yaitu admin dan mahasiswa. API terdiri dari 10 endpoint dan dibangun dengan Go, Fiber v2, GORM, PostgreSQL, JWT, dan bcrypt. Cara menjalankan proyek dijelaskan pada file README.md di repository.

Langkah-langkah:

1. Hasil menjalankan migration database

```
PS D:\PBL PRAK\api-students\UTS> go run .
2026/10/08 01:31:09 Migrasi selesai
PS D:\PBL PRAK\api-students\UTS>
```

Migration dijalankan melalui perintah `go run .` pada folder UTS. Program membaca konfigurasi dari file `.env`, terhubung ke database PostgreSQL `db_siakad`, lalu menjalankan AutoMigrate GORM untuk membuat tabel `users`, `students`, `courses`, dan `enrollments`. Pesan "Migrasi selesai" menunjukkan seluruh proses berjalan tanpa error.

2. Daftar tabel pada database db_siakad

```
PS D:\PBL PRAK\api-students\UTS> psql -U postgres -d db_siakad -c "\dt"
Password for user postgres:
List of tables
 Schema |   Name   | Type  | Owner
-----+-----+-----+-----
 public | courses  | table | postgres
 public | enrollments | table | postgres
 public | students | table | postgres
 public | users    | table | postgres
(4 rows)
```

Pengecekan melalui psql menunjukkan bahwa keempat tabel yang dibutuhkan, yaitu users, students, courses, dan enrollments, sudah terbentuk di database db_siakad. Hal ini sesuai dengan model data minimal yang diminta pada soal.

3. Struktur tabel enrollments

```
PS D:\PBL PRAK\api-students\UTS> psql -U postgres -d db_siakad -c "\d enrollments"
Password for user postgres:
      Column      |      Type      |      Collation      | Nullable |      Default
-----+-----+-----+-----+-----+-----
id                | bigint         |                     | not null | nextval('enrollments_id_seq'::regclass)
student_id        | bigint         |                     | not null |
course_id         | bigint         |                     | not null |
tahun_akademik    | character varying(25) |                     | not null |
created_at        | timestamp with time zone |                     |          |
Indexes:
    "enrollments_pkey" PRIMARY KEY, btree (id)
    "idx_enroll_unique" UNIQUE, btree (student_id, course_id, tahun_akademik)
Foreign-key constraints:
    "fk_courses_enrollments" FOREIGN KEY (course_id) REFERENCES courses(id)
    "fk_students_enrollments" FOREIGN KEY (student_id) REFERENCES students(id)
```

Tabel enrollments memiliki kolom id, student_id, course_id, tahun_akademik, dan created_at. Terdapat unique index idx_enroll_unique pada kombinasi student_id, course_id, dan tahun_akademik sehingga mahasiswa tidak dapat mengambil mata kuliah yang sama dua kali pada tahun akademik yang sama. Selain itu, student_id dan course_id terhubung sebagai foreign key ke tabel students dan courses.

4. Hasil menjalankan seeder

```
PS D:\PBL PRAK\api-students\UTS> go get golang.org/x/crypto/bcrypt
go: added golang.org/x/crypto v0.57.0
PS D:\PBL PRAK\api-students\UTS> go mod tidy
PS D:\PBL PRAK\api-students\UTS> go run .
2026/10/08 01:46:25 Migrasi selesai
2026/10/08 01:46:26 Seeder selesai
PS D:\PBL PRAK\api-students\UTS>
```

Seeder dijalankan setelah migration melalui perintah go run . Seeder membuat 1 akun admin, 20 akun mahasiswa beserta data profilnya, dan 10 mata kuliah dalam satu transaction. Pesan "Seeder selesai" menunjukkan seluruh data berhasil dimasukkan.

5. Jumlah data hasil seeding

```
PS D:\PBL PRAK\api-students\UTS> psql -U postgres -d db_siakad -c "SELECT (SELECT COUNT(*) FROM users) AS users, (SELECT COUNT(*) FROM students) AS students, (SELECT COUNT(*) FROM courses) AS courses;"
Password for user postgres:
 users | students | courses
-----+-----+-----
    21 |        20 |        10
(1 row)
```

Query jumlah data menunjukkan terdapat 21 baris pada tabel users (1 admin dan 20 mahasiswa), 20 baris pada tabel students, dan 10 baris pada tabel courses. Jumlah ini memenuhi ketentuan seeder minimal pada soal.

6. Password tersimpan dalam bentuk hash

```
PS D:\PBL PRAK\api-students\UTS> psql -U postgres -d db_siakad -c "SELECT id, email, role, LEFT(password, 25) AS password_hash FROM users LIMIT 5;"
Password for user postgres:
id | email | role | password_hash
---+-----+-----+-----
 1 | admin@siakad.test | admin | $2a$10$XKn.2GAcZfp0ghJiRz
 2 | mhs01@siakad.test | mahasiswa | $2a$10$GG3mAJuBJ9XD88/0xj
 3 | mhs02@siakad.test | mahasiswa | $2a$10$zxtMMqXQ0wXE8cRAVI
 4 | mhs03@siakad.test | mahasiswa | $2a$10$nnrN/U5Aqd.a0xTLHa
 5 | mhs04@siakad.test | mahasiswa | $2a$10$M5wHhvx8xl4t7Ikaz
(5 rows)
```

Kolom password pada tabel users tidak menyimpan teks asli, melainkan hasil hash bcrypt (diawali \$2a\$10\$). Dengan demikian ketentuan bahwa password harus disimpan dalam bentuk hash terpenuhi.

7. Hasil menjalankan server

```
PS D:\PBL PRAK\api-students\UTS> go run .
2026/10/08 02:29:16 Migrasi selesai
2026/10/08 02:29:16 Seeder dilewati: data sudah ada
2026/10/08 02:29:16 Seeder selesai

Fiber v2.52.15
http://127.0.0.1:3000
(bound on host 0.0.0.0 and port 3000)

Handlers ..... 2 Processes ..... 1
Prefork ..... Disabled PID ..... 10752
```

Server dijalankan melalui perintah `go run .`. Aplikasi melakukan koneksi ke database dan menjalankan migration, sedangkan seeder dilewati karena data sudah tersedia. Setelah itu Fiber berjalan pada port 3000 dan siap menerima request.

8. Format response sukses

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/ping
HTTP/1.1 200 OK
Date: Wed, 07 Oct 2026 19:35:15 GMT
Content-Type: application/json
Content-Length: 65

{"data":null,"message":"API SIAKAD Mini berjalan","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Request ke endpoint `/api/v1/ping` mengembalikan status 200 OK dengan body JSON yang berisi success, message, dan data. Format response ini seragam dan digunakan oleh seluruh endpoint API.

Catatan: endpoint `/api/v1/ping` hanya digunakan pada tahap awal untuk menguji server dan format response. Endpoint ini sudah dihapus karena soal mewajibkan autentikasi token pada semua endpoint selain login.

9. Format response error

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/tidak-ada
HTTP/1.1 404 Not Found
Date: Wed, 07 Oct 2026 19:37:26 GMT
Content-Type: application/json
Content-Length: 58

{"message":"Cannot GET /api/v1/tidak-ada","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Request ke alamat yang tidak tersedia mengembalikan status 404 Not Found dengan body JSON berisi success bernilai false dan pesan error. Seluruh error ditangani oleh error handler terpusat sehingga formatnya konsisten dan detail internal tidak ditampilkan ke pengguna.

10. Login Berhasil

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/auth/login -H "Content-Type: application/json" --data "@$env:TEMP\login.json"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 03:08:21 GMT
Content-Type: application/json
Content-Length: 334
X-Ratelimit-Limit: 5
X-Ratelimit-Remaining: 5
X-Ratelimit-Reset: 60

{"data":{"access_token":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoxLCJyb2xlIjoiaWVWRTAw4iLCJleHAiOjE3OTE1MTUzMDU5Imh0dCI6MTc5MTQyODkwMX0.wCn1TrmXHO4YgKfKQCdOP7Ytgva20LHnPO0ZYus-pPg","expires_in":86400,"token_type":"Bearer","user":{"email":"admin@siakad.test","id":1,"role":"admin"}}, "message":"Login berhasil","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Request POST ke /api/v1/auth/login dengan kredensial yang benar mengembalikan status 200 beserta access_token JWT, tipe token Bearer, masa berlaku (expires_in), dan data user yang terdiri dari id, email, dan role. Password tidak ikut dikembalikan pada response.

11. Login Gagal

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/auth/login -H "Content-Type: application/json" --data "@$env:TEMP\salah.json"
HTTP/1.1 401 Unauthorized
Date: Thu, 08 Oct 2026 03:37:14 GMT
Content-Type: application/json
Content-Length: 55
X-Ratelimit-Limit: 5
X-Ratelimit-Remaining: 5
X-Ratelimit-Reset: 60

{"message":"Email atau password salah","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Login dengan password yang salah ditolak dengan status 401 Unauthorized dan body JSON berisi success bernilai false. Pesan error dibuat sama untuk email yang tidak terdaftar maupun password yang salah, sehingga tidak membocorkan informasi apakah suatu email terdaftar di sistem.

12. Akses /auth/me dengan token

```
PS D:\PBL PRAK\api-students\UTS> $token = (curl.exe -s -X POST http://localhost:3000/api/v1/auth/login -H "Content-Type: application/json" --data "@$env:TEMP\login.json" | ConvertFrom-Json).data.access_token
>> curl.exe -i http://localhost:3000/api/v1/auth/me -H "Authorization: Bearer $token"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 03:38:38 GMT
Content-Type: application/json
Content-Length: 110

{"data":{"user":{"email":"admin@siakad.test","id":1,"role":"admin"}}, "message":"Data pengguna", "success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Endpoint /api/v1/auth/me yang dilindungi middleware autentikasi dapat diakses setelah mengirim token JWT pada header Authorization dengan format Bearer. Response berisi data pengguna yang sedang login.

13. Akses /auth/me tanpa token

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/auth/me
HTTP/1.1 401 Unauthorized
Date: Wed, 07 Oct 2026 19:59:01 GMT
Content-Type: application/json
Content-Length: 69

{"message":"Token tidak ditemukan atau format salah","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Request ke endpoint yang dilindungi tanpa menyertakan token ditolak oleh middleware dengan status 401 Unauthorized, sehingga data pengguna tidak dapat diakses tanpa autentikasi.

14. Pembatasan percobaan login (rate limit)

```
PS D:\PBL PRAK\api-students\UTS> 1..6 | ForEach-Object { curl.exe -s -o NUL -w "percobaan $_ : HTTP %
{http_code}\n" -X POST http://localhost:3000/api/v1/auth/login -H "Content-Type: application/json" --
data "@$env:TEMP\salah.json" }
percobaan 1 : HTTP 401
percobaan 2 : HTTP 401
percobaan 3 : HTTP 401
percobaan 4 : HTTP 401
percobaan 5 : HTTP 401
percobaan 6 : HTTP 429
PS D:\PBL PRAK\api-students\UTS>
```

Endpoint login dibatasi maksimal 5 request per menit untuk setiap alamat IP guna mencegah serangan brute force. Request yang melebihi batas ditolak dengan status 429 Too Many Requests.

15. Validasi login gagal

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/auth/login -H "Content-Type: applicati
on/json" --data "@$env:TEMP\val1.json"
HTTP/1.1 422 Unprocessable Entity
Date: Thu, 08 Oct 2026 03:44:29 GMT
Content-Type: application/json
Content-Length: 126
X-Ratelimit-Limit: 5
X-Ratelimit-Remaining: 5
X-Ratelimit-Reset: 60

{"errors":{"email":["Format email tidak valid"],"password":["minimal 8 karakter"]}, "message":"Validasi gagal", "success":
false}
PS D:\PBL PRAK\api-students\UTS>
```

Login dengan email berformat salah dan password kurang dari 8 karakter ditolak dengan status 422 Unprocessable Entity. Response memuat objek errors yang merinci pesan kesalahan untuk setiap field, sehingga klien dapat mengetahui bagian mana yang harus diperbaiki.

16. Daftar mahasiswa dengan pagination

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i "http://localhost:3000/api/v1/students?page=1&per_page=3" -H "Authorization: Bearer $admin"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 03:55:06 GMT
Content-Type: application/json
Content-Length: 514

{"data":[{"id":1,"user_id":2,"nim":"434241000001","nama":"Rina Putri","prodi":"Teknik Informatika","angkatan":2022,"ipk_terakhir":3.85},{"id":2,"user_id":3,"nim":"434241000002","nama":"Budi Santoso","prodi":"Sistem Informasi","angkatan":2023,"ipk_terakhir":3.62},{"id":3,"user_id":4,"nim":"434241000003","nama":"Citra Lestari","prodi":"Teknik Komputer","angkatan":2024,"ipk_terakhir":3.45}], "message":"Data mahasiswa berhasil diambil", "meta":{"current_page":1,"per_page":3,"total":20,"last_page":7},"success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Endpoint GET /api/v1/students dapat diakses admin dan mengembalikan daftar mahasiswa dengan pagination. Objek meta memuat current_page, per_page, total, dan last_page.

17. Filter dan pengurutan

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i "http://localhost:3000/api/v1/students?angkatan=2022&sort=-ipk_terakhir&per_page=5" -H "Authorization: Bearer $admin"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 03:59:54 GMT
Content-Type: application/json
Content-Length: 772

{"data":[{"id":1,"user_id":2,"nim":"434241000001","nama":"Rina Putri","prodi":"Teknik Informatika","angkatan":2022,"ipk_terakhir":3.85},{"id":9,"user_id":10,"nim":"434241000009","nama":"Indah Sari","prodi":"Teknik Komputer","angkatan":2022,"ipk_terakhir":3.5},{"id":5,"user_id":6,"nim":"434241000005","nama":"Eka Wulandari","prodi":"Sistem Informasi","angkatan":2022,"ipk_terakhir":3.15},{"id":13,"user_id":14,"nim":"434241000013","nama":"Maya Anggraini","prodi":"Teknik Informatika","angkatan":2022,"ipk_terakhir":2.75},{"id":17,"user_id":18,"nim":"434241000017","nama":"Qonita Aulia","prodi":"Sistem Informasi","angkatan":2022,"ipk_terakhir":2.7}], "message":"Data mahasiswa berhasil diambil", "meta":{"current_page":1,"per_page":5,"total":5,"last_page":1},"success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Parameter angkatan menyaring data dan sort=-ipk_terakhir mengurutkan dari IPK tertinggi. Parameter prodi dan search juga tersedia.

18. Akses daftar oleh non-admin (403)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/students -H "Authorization: Bearer $mhs"
HTTP/1.1 403 Forbidden
Date: Thu, 08 Oct 2026 04:01:37 GMT
Content-Type: application/json
Content-Length: 71

{"message":"Anda tidak memiliki akses ke resource ini","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Akun mahasiswa yang mengakses daftar mahasiswa ditolak dengan status 403 karena endpoint ini hanya untuk admin.

19. Tambah mahasiswa (201)

```
PS D:\PBL PRAK\api-students\UTS> '{ "nim": "187221009999", "nama": "Budi Uji", "email": "budi.uji@siakad.test", "prodi": "Sistem Informasi", "angkatan": 2024, "ipk_terakhir": 3.2 }' | Out-File -Encoding ascii $env:TEMP\new.json
>> curl.exe -i -X POST http://localhost:3000/api/v1/students -H "Content-Type: application/json" -H "Authorization: Bearer $admin" --data "@$env:TEMP\new.json"
HTTP/1.1 201 Created
Date: Thu, 08 Oct 2026 04:03:10 GMT
Content-Type: application/json
Content-Length: 221

{"data": {"angkatan": 2024, "email": "budi.uji@siakad.test", "id": 21, "ipk_terakhir": 3.2, "nama": "Budi Uji", "nim": "187221009999", "prodi": "Sistem Informasi", "user_id": 22}, "message": "Mahasiswa berhasil ditambahkan", "success": true}
PS D:\PBL PRAK\api-students\UTS>
```

Admin menambahkan mahasiswa baru dan sistem membuat akun user (role mahasiswa, password awal berupa NIM yang di-hash) beserta data mahasiswa dalam satu transaction. Response berstatus 201 Created.

20. Data duplikat (422)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/students -H "Content-Type: application/json" -H "Authorization: Bearer $admin" --data "@$env:TEMP\new.json"
HTTP/1.1 422 Unprocessable Entity
Date: Thu, 08 Oct 2026 04:07:35 GMT
Content-Type: application/json
Content-Length: 119

{"errors": {"email": ["Email sudah terdaftar"], "nim": ["NIM sudah terdaftar"]}, "message": "Validasi gagal", "success": false}
PS D:\PBL PRAK\api-students\UTS>
```

Menambahkan data dengan NIM dan email yang sudah terdaftar ditolak dengan status 422. Response memuat pesan kesalahan per field.

21. Hak akses mahasiswa pada detail (200 lalu 403)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/students/$id1 -H "Authorization: Bearer $mhs"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 04:09:25 GMT
Content-Type: application/json
Content-Length: 242

{"data": {"angkatan": 2022, "batas_sks": 24, "enrollments": [], "id": 1, "ipk_terakhir": 3.85, "nama": "Rina Putri", "nim": "43424100001", "prodi": "Teknik Informatika", "total_sks": 0, "user_id": 2}, "message": "Detail mahasiswa berhasil diambil", "success": true}
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/students/$id2 -H "Authorization: Bearer $mhs"
HTTP/1.1 403 Forbidden
Date: Thu, 08 Oct 2026 04:09:33 GMT
Content-Type: application/json
Content-Length: 75

{"message": "Anda hanya dapat mengakses data milik sendiri", "success": false}
PS D:\PBL PRAK\api-students\UTS>
```

Mahasiswa dapat melihat data miliknya sendiri (200), tetapi ditolak dengan status 403 saat mengakses data mahasiswa lain sehingga data pribadi tetap terlindungi.

22. Perbarui data mahasiswa

```
PS D:\PBL PRAK\api-students\UTS> '{ "nama": "Budi Uji Update", "prodi": "Informatika", "angkatan": 2023, "ipk_terakhir": 3.6 }' | Out-File -Encoding ascii $env:TEMP\upd.json
>> curl.exe -i -X PUT http://localhost:3000/api/v1/students/$newId -H "Content-Type: application/json" -H "Authorization: Bearer $admin" --data "@$env:TEMP\upd.json"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 04:11:06 GMT
Content-Type: application/json
Content-Length: 196

{"data": {"id": 21, "user_id": 22, "nim": "187221009999", "nama": "Budi Uji Update", "prodi": "Informatika", "angkatan": 2023, "ipk_terakhir": 3.6}, "message": "Data mahasiswa berhasil diperbarui", "success": true}
PS D:\PBL PRAK\api-students\UTS>
```

Admin memperbarui nama, prodi, angkatan, dan IPK mahasiswa. NIM tidak dapat diubah.

23. Soft delete mahasiswa

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X DELETE http://localhost:3000/api/v1/students/$newid -H "Authorization: Bearer $admin"
HTTP/1.1 204 No Content
Date: Thu, 08 Oct 2026 04:12:58 GMT

PS D:\PBL PRAK\api-students\UTS>
```

Penghapusan mahasiswa dilakukan secara soft delete dengan mengisi kolom `deleted_at`. Response berstatus 204 tanpa body.

24. Mahasiswa terhapus: data tidak bisa diakses, akun tidak bisa login

```
HTTP/1.1 404 Not Found
Date: Thu, 08 Oct 2026 04:14:31 GMT
Content-Type: application/json
Content-Length: 55

{"message":"Mahasiswa tidak ditemukan","success":false}HTTP/1.1 401 Unauthorized
Date: Thu, 08 Oct 2026 04:14:31 GMT
Content-Type: application/json
Content-Length: 55
X-Ratelimit-Limit: 5
X-Ratelimit-Remaining: 5
X-Ratelimit-Reset: 59

{"message":"Email atau password salah","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Setelah di-soft delete, data mahasiswa tidak lagi dapat diakses (404) dan akunnya tidak dapat digunakan untuk login (401).

25. Daftar mata kuliah (200, ada terisi dan sisa_kuota)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/courses -H "Authorization: Bearer $mhs"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 04:44:18 GMT
Content-Type: application/json
Content-Length: 1226

{"data":[{"id":1,"kode_mk":"IF101","nama_mk":"Algoritma dan Pemrograman","sks":3,"semester":1,"kuota":30,"terisi":0,"sisa_kuota":30},{"id":2,"kode_mk":"IF102","nama_mk":"Basis Data","sks":3,"semester":2,"kuota":30,"terisi":0,"sisa_kuota":30},{"id":3,"kode_mk":"IF103","nama_mk":"Struktur Data","sks":3,"semester":2,"kuota":30,"terisi":0,"sisa_kuota":30},{"id":4,"kode_mk":"IF104","nama_mk":"Jaringan Komputer","sks":3,"semester":3,"kuota":25,"terisi":0,"sisa_kuota":25},{"id":5,"kode_mk":"IF105","nama_mk":"Pemrograman Web","sks":4,"semester":3,"kuota":25,"terisi":0,"sisa_kuota":25},{"id":6,"kode_mk":"IF106","nama_mk":"Rekayasa Perangkat Lunak","sks":3,"semester":3,"kuota":25,"terisi":0,"sisa_kuota":25},{"id":7,"kode_mk":"IF107","nama_mk":"Pemrograman Back End","sks":4,"semester":4,"kuota":2,"terisi":0,"sisa_kuota":2},{"id":8,"kode_mk":"IF108","nama_mk":"Keamanan Siber","sks":3,"semester":5,"kuota":2,"terisi":0,"sisa_kuota":2},{"id":9,"kode_mk":"IF109","nama_mk":"Kecerdasan Buatan","sks":3,"semester":5,"kuota":20,"terisi":0,"sisa_kuota":20},{"id":10,"kode_mk":"IF110","nama_mk":"Etika Profesi","sks":2,"semester":6,"kuota":30,"terisi":0,"sisa_kuota":30}],message:"Data mata kuliah berhasil diambil","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Endpoint `GET /api/v1/courses` dapat diakses semua role. Setiap mata kuliah menyertakan jumlah terisi dan sisa kuota yang dihitung dari tabel enrollments.

26. Filter semester dan ketersediaan

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i "http://localhost:3000/api/v1/courses?semester=1&available=true" -H "Authorization: Bearer $mhs"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 04:45:51 GMT
Content-Type: application/json
Content-Length: 195

{"data":[{"id":1,"kode_mk":"IF101","nama_mk":"Algoritma dan Pemrograman","sks":3,"semester":1,"kuota":30,"terisi":0,"sisa_kuota":30}],message:"Data mata kuliah berhasil diambil","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Parameter semester menyaring mata kuliah per semester, sedangkan available=true hanya menampilkan mata kuliah yang kuotanya belum penuh.

27. Mengambil mata kuliah (201)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/enrollments -H "Content-Type: application/json" -H "Authorization: Bearer $mhs" --data "@$env:TEMP\ei.json"
HTTP/1.1 201 Created
Date: Thu, 08 Oct 2026 04:47:15 GMT
Content-Type: application/json
Content-Length: 302

{"data":{"batas_sks":24,"course":{"kode_mk":"IF101","nama_mk":"Algoritma dan Pemrograman","sks":3},"course_id":1,"created_at":"2026-10-08T11:47:15.8917919+07:00","id":1,"student_id":1,"tahun_akademik":"2026/2027-Ganjil","total_sks":3},"message":"Mata kuliah berhasil ditambahkan ke KRS","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Mahasiswa menambahkan mata kuliah ke KRS. Pengecekan duplikasi, kuota, dan batas SKS dilakukan dalam satu transaction. Response berstatus 201 beserta total SKS dan batas SKS.

28. Mata kuliah yang sama (409)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/enrollments -H "Content-Type: application/json" -H "Authorization: Bearer $mhs" --data "@$env:TEMP\ei.json"
HTTP/1.1 409 Conflict
Date: Thu, 08 Oct 2026 04:48:41 GMT
Content-Type: application/json
Content-Length: 96

{"message":"Mata kuliah ini sudah pernah diambil pada tahun akademik yang sama","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Mata kuliah yang sama pada tahun akademik yang sama ditolak dengan status 409 Conflict.

29. Validasi tahun akademik (422)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/enrollments -H "Content-type: application/json" -H "Authorization: Bearer $mhs" --data "@$env:TEMP\ebad.json"
HTTP/1.1 422 Unprocessable Entity
Date: Thu, 08 Oct 2026 04:50:11 GMT
Content-Type: application/json
Content-Length: 114

{"errors":{"tahun_akademik":["format harus seperti 2026/2027-Ganjil"]},"message":"Validasi gagal","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Format tahun akademik yang salah ditolak dengan status 422 beserta pesan format yang benar.

30. Admin mengambil mata kuliah (403)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/enrollments -H "Content-Type: application/json" -H "Authorization: Bearer $admin" --data "@$env:TEMP\ei.json"
HTTP/1.1 403 Forbidden
Date: Thu, 08 Oct 2026 04:52:40 GMT
Content-Type: application/json
Content-Length: 71

{"message":"Anda tidak memiliki akses ke resource ini","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Pengambilan mata kuliah hanya untuk role mahasiswa, sehingga akun admin ditolak dengan status 403.

31. Batas SKS terlampaui

```
PS D:\PBL PRAK\api-students\UTS> 2.10 | ForEach-Object {
>>   ["course_id":' + $_ + ', "tahun_akademik": "2026/2027-Ganjil"]' | Out-File -Encoding ascii $env:TEMP\e.json
>>   $r = curl.exe -s -X POST http://localhost:3000/api/v1/enrollments -H "Content-Type: application/json" -H "Authorization: Bearer $mhs" --data "@$env:TEMP\e.json" | ConvertFrom-Json
>>   "course $_ : success= $($r.success) | $($r.message)"
>> }
course 2 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 3 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 4 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 5 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 6 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 7 : success=True | Mata kuliah berhasil ditambahkan ke KRS
course 8 : success=False | Total SKS melebihi batas 24 SKS (IPK 3.85). SKS terpakai 23, sisa 1 SKS, sedangkan mata kuliah ini 3 SKS
course 9 : success=False | Total SKS melebihi batas 24 SKS (IPK 3.85). SKS terpakai 23, sisa 1 SKS, sedangkan mata kuliah ini 3 SKS
course 10 : success=False | Total SKS melebihi batas 24 SKS (IPK 3.85). SKS terpakai 23, sisa 1 SKS, sedangkan mata kuliah ini 2 SKS
PS D:\PBL PRAK\api-students\UTS>
```

Mahasiswa mengambil mata kuliah satu per satu sampai total SKS melampaui batas sesuai IPK. Pengambilan ditolak dengan status 422 dan pesan yang menyebut batas, SKS terpakai, dan sisa SKS.

32. Detail mahasiswa dengan KRS

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i http://localhost:3000/api/v1/students/$id1 -H "Authorization: Bearer $admin"
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 04:55:48 GMT
Content-Type: application/json
Content-Length: 1784

{"data":{"angkatan":2022,"batas_sks":24,"enrollments":[{"id":1,"student_id":1,"course_id":1,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:47:15.891791+07:00","course":{"id":1,"kode_mk":"IF101","nama_mk":"Algoritma dan Pemrograman","sks":3,"semester":1,"kuota":30}},{"id":2,"student_id":1,"course_id":2,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:19.3892+07:00","course":{"id":2,"kode_mk":"IF102","nama_mk":"Basis Data","sks":3,"semester":2,"kuota":30}},{"id":3,"student_id":1,"course_id":3,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:19.639723+07:00","course":{"id":3,"kode_mk":"IF103","nama_mk":"Struktur Data","sks":3,"semester":2,"kuota":30}},{"id":4,"student_id":1,"course_id":4,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:19.920242+07:00","course":{"id":4,"kode_mk":"IF104","nama_mk":"Jaringan Komputer","sks":3,"semester":3,"kuota":25}},{"id":5,"student_id":1,"course_id":5,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:20.239568+07:00","course":{"id":5,"kode_mk":"IF105","nama_mk":"Pemrograman Web","sks":4,"semester":3,"kuota":25}},{"id":6,"student_id":1,"course_id":6,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:20.522008+07:00","course":{"id":6,"kode_mk":"IF106","nama_mk":"Rekayasa Perangkat Lunak","sks":3,"semester":4,"kuota":25}},{"id":7,"student_id":1,"course_id":7,"tahun_akademik":"2026/2027-Ganjil","created_at":"2026-10-08T11:54:20.764708+07:00","course":{"id":7,"kode_mk":"IF107","nama_mk":"Pemrograman Back End","sks":4,"semester":4,"kuota":25}}],"id":1,"ipk_terakhir":3.85,"nama":"Rina Putri","nim":"434241000001","prodi":"Teknik Informatika","total_sks":23,"user_id":2,"message":"Detail mahasiswa berhasil diambil","success":true}
PS D:\PBL PRAK\api-students\UTS>
```

Detail mahasiswa memuat data profil, daftar mata kuliah yang diambil pada KRS, total SKS yang dihitung dari mata kuliah tersebut, dan batas SKS sesuai IPK terakhir.

33. Kuota penuh (422)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X POST http://localhost:3000/api/v1/enrollments -H "Content-Type: application/json" -H "Authorization: Bearer $mhs2" --data "@$env:TEMP\el.json"
HTTP/1.1 422 Unprocessable Entity
Date: Thu, 08 Oct 2026 04:57:38 GMT
Content-Type: application/json
Content-Length: 59

{"message":"Kuota mata kuliah sudah penuh","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Mata kuliah yang kuotanya sudah penuh tidak dapat diambil mahasiswa lain. Permintaan ditolak dengan status 422. Kuota diatur menjadi 1 hanya untuk keperluan pengujian.

34. Filter available=true setelah kuota penuh

```
PS D:\PBL PRAK\api-students\UTS> (curl.exe -s "http://localhost:3000/api/v1/courses?available=true" -H "Authorization: Bearer $mhs" | ConvertFrom-Json).data | Select-Object id, kode_mk, kuota, terisi, sisa_kuota | Format-Table
id kode_mk kuota terisi sisa_kuota
-----
2 IF102 30 1 29
3 IF103 30 1 29
4 IF104 25 1 24
5 IF105 25 1 24
6 IF106 25 1 24
7 IF107 2 1 1
8 IF108 2 0 2
9 IF109 20 0 20
10 IF110 30 0 30
PS D:\PBL PRAK\api-students\UTS>
```

Mata kuliah dengan kuota penuh tidak muncul saat parameter available=true digunakan.

35. Membatalkan KRS orang lain (403) dan yang tidak ada (404)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X DELETE http://localhost:3000/api/v1/enrollments/$enrId -H "Authorization: Bearer $mhs2"
>> curl.exe -i -X DELETE http://localhost:3000/api/v1/enrollments/99999 -H "Authorization: Bearer $mhs"
HTTP/1.1 403 Forbidden
Date: Thu, 08 Oct 2026 05:01:01 GMT
Content-Type: application/json
Content-Length: 76

{"message":"Anda hanya dapat membatalkan KRS milik sendiri","success":false}HTTP/1.1 404 Not Found
Date: Thu, 08 Oct 2026 05:01:02 GMT
Content-Type: application/json
Content-Length: 54

{"message":"Data KRS tidak ditemukan","success":false}
PS D:\PBL PRAK\api-students\UTS>
```

Mahasiswa yang membatalkan KRS milik mahasiswa lain ditolak dengan status 403, sedangkan ID yang tidak ada mengembalikan status 404.

36. Membatalkan KRS sendiri (204)

```
PS D:\PBL PRAK\api-students\UTS> curl.exe -i -X DELETE http://localhost:3000/api/v1/enrollments/$enrId -H "Authorization: Bearer $mhs"
HTTP/1.1 204 No Content
Date: Thu, 08 Oct 2026 05:13:26 GMT
PS D:\PBL PRAK\api-students\UTS>
```

Mahasiswa membatalkan mata kuliah dari KRS miliknya. Response berstatus 204 dan kuota mata kuliah kembali bertambah karena dihitung dari tabel enrollments.

Kesimpulan

Seluruh 10 endpoint SIAKAD Mini API telah diimplementasikan dan diuji, mencakup autentikasi JWT, pembatasan akses berdasarkan role, validasi input, serta business rule batas SKS, duplikasi, dan kuota mata kuliah. Status 500 tidak didemonstrasikan karena tidak ada kondisi yang dapat memicunya secara wajar. Meski begitu, error tak terduga ditangani oleh error handler terpusat dan pada mode production pesannya tidak membocorkan detail internal. Kode lengkap dan riwayat commit dapat dilihat pada repository: <https://github.com/finarch1ve/api-students/tree/main/UTS>